INCH-POUND
MIL-T-82456A(OS)
26 August 1988
SUPERSEDING
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10 May 1966 and
Amendment 1
12 March 1979

MILITARY SPECIFICATION

TEST SET, GYROSCOPE, MARK 484 MOD 1
(For Torpedo Mk 46 Mod 0 through 5)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Test Set, Gyroscope, Mark 484 Mod 1, hereinafter referred to as the "test set."

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified (see 6.2), the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-B-636	Boxes, Shipping, Fiberboard

MILITARY

MIL-P-116	Preservation, Methods of
MIL-S-901	Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for
MIL-E-16400	Electronic, Interior Communication and Navigation Equipment, Naval Ship and Shore, General Specification for

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-167-1	Mechanical Vibration of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, Maryland 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4921

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (Attn: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified (see 6.2), the issues shall be those cited in the solicitation.

DRAWINGS

NAVAL SEA SYSTEMS COMMAND (CAGE Code Ident 10001)

PL3277490	Test Set, Gyroscope, Mk 484 Mod 1
LD541118	Course Gyroscope
Dwg.2557813	Course Gyroscope

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

2.2 Non-Government publications. The following document forms a part of this document to the extent specified herein. Unless otherwise specified (see 6.2), the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS are the issues of the documents cited in the solicitation.

PUBLICATIONS

UNIFORM CLASSIFICATION COMMITTEE AGENT

Uniform Freight Classification Rules and Regulations

(Application for copies should be addressed to the Uniform Classification Committee Agent, Traiff Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

(Non-Government standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of Precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Basic requirements.

3.2.1 Materials and treatments of materials and parts. Unless otherwise specified in the contract or purchase order (see 6.2), all materials and treatments of materials and parts used in the manufacture of the test set shall be in accordance with the requirements specified herein and with the applicable documents listed on PL3277490.

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3.2.2 Gauges. The contractor shall provide whatever gauges are necessary and adequate to assure that the material and parts of the test set will meet the dimensions and requirements shown on applicable documents. When requested by the contractor, the contracting activity will furnish existing pertinent Navy drawing of final acceptance gauges for use only in designing of the contractor's gauges. This action does not relieve the contractor of the responsibility for designing and manufacturing the gauges needed for satisfactory fulfillment of the contract requirements.

3.2.3 Inspection and test equipment facilities. The contractor shall furnish and maintain all necessary inspection and test equipment and facilities and provide personnel for performing all quality conformance tests. The test equipment shall be adequate to permit performance of the required quality conformance tests for the test set.

3.3 Performance requirements and physical characteristics.

3.3.1 Power supply. The test set shall meet the performance requirements of 3.3.2, 3.3.3, and 3.3.4 using a self-contained battery.

3.3.2 Output. The maximum output current of the test set shall be 10.0 milliamperes (ma) direct current (dc).

3.3.3 Resistance measurements. With the switches positioned as shown in table I, and the test switch depressed, the ohmmeter shall measure any resistance across the specified pins within the range indicated. The center scale accuracy of the test set shall be ±10 percent.

TABLE I. Switch positions.

Test selector switch positions	Center scale sensitivity (Ohms)	Resistance range (Ohms)	Location of resistance load connector J1
Ins	2,500	0 to 60,000	L, H, E; and B to male guide pin
Squib	2.5	0 to 60	P to L
Cont	2.5	0 to 60	J to H
Motor	250	0 to 6000	D to E
Pot	2500	0 to 60,000	A to B
Pot	2500	0 to 60,000	A to C
Ext	2.5 ohms to 2500 ohms in decade steps indicated by ohms center scale switch position	Four ranges: 0 to 60 0 to 600 0 to 6000 0 to 60,000	Test plugs P1 to P2

3.3.4 Calibration. When the test selector switch is in the calibration position, rotation of the calibration adjust control shall allow adjustment of the ohmmeter needle to the infinity mark on the face of the meter.

3.4 Environmental requirements. Unless otherwise specified (see 6.2), the test set shall conform to the performance requirements and physical characteristics of this specification during and after exposure to the following environments.

3.4.1 Temperature. The test set shall conform to the temperature requirements of MIL-E-16400, Range 4.

3.4.2 Humidity. The test set shall conform to the humidity requirements of MIL-E-16400 (see 6.4.1).

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3.4.3 Shock. The test set shall conform to the shock requirements of MIL-S-901, grade A, type A, lightweight equipment, except that the test set shall be in the nonoperating condition, and the hammer drops shall be from 1 foot, 1½ feet, and 3 feet (see 6.4.2).

3.4.4 Vibration. The test set shall conform to the vibration requirements of MIL-STD-167-1, except that the variable frequency and endurance vibration requirements are replaced by the following frequency requirements (see 3.4.4.1 and 6.4.3).

3.4.4.1 Vibration frequency (variable). The test set, in the operating condition, shall withstand vibration from 5 to 15 Hertz (Hz) at an amplitude of 0.030 ± 0.006 inch, and from 16 to 25 Hz at an amplitude of 0.020 ± 0.004 inch. The vibration shall be applied in discrete frequency intervals of 1 Hz, and shall be applied for 1 minute at each interval.

3.4.5 Inclination. The test set shall conform to the inclination requirements of MIL-E-16400.

3.5 Service life. The service life of the test set (less battery) shall be 10,000 hours of operation.

3.6 Identification. The test set shall be permanently identified in accordance with MIL-STD-130 and the applicable documents listed in PL3277490.

3.7 Workmanship. The workmanship and finish shall be of sufficiently high grade to assure satisfactory operation consistent with the requirements of this specification. All parts used in the assembly shall be thoroughly clean and free of excess materials, chips, and loose or splattered foreign materials. The standard of workmanship exhibited in any approved first article sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.5)

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4.3 Environmental test conditions. Unless otherwise specified (see 6.2), the test sets shall be subjected to tests under the following conditions:

a. Temperature	60 to 90°F (room ambient)
b. Relative Humidity	Room ambient, up to 95 percent
c. Altitude	Normal ground atmospheric pressure

4.4 First article test and inspection.

4.4.1 First article sample. Prior to the start of production, a sample of one test set, manufactured in accordance with 3.1, shall be subjected to the tests of 4.4.2 through 4.4.3 and 4.6 at a facility designated by the contracting activity. Acceptance of this sample shall be based on no defects or failures of the sample. Production of the test set by the contractor, prior to acceptance of the first article sample, shall be at the contractor's risk.

4.4.2. Environmental tests. The test set shall be subjected to the following environmental tests. Unless otherwise specified, inability of the test set to pass the quality conformance tests and inspections of 4.6, subsequent to the environmental tests, shall constitute a failure.

4.4.2.1. Temperature. Mount the test set in a temperature-humidity test chamber.

a. Not-operating condition. Subject the test set, in the not-operating condition, to the nonoperating condition temperature test specified in MIL-E-16400.
b. Operating condition. Subject the test set, in the operating condition, to the operating condition, range 4, temperature test specified in MIL-E-16400.

4.4.2.2 Humidity. Mount the test set in the temperature-humidity chamber and subject it to the humidity test specified in MIL-E-16400.

4.4.2.3 Shock. Mount the test set as specified in the shock test in MIL-E-16400, and secure it to the shock test fixture. Subject the test set to the shock tests specified in MIL-S-901, grade A, type A, lightweight equipment, except that the test set shall be not-operating and the hammer drop heights shall be 1, 1½, and 3 feet.

4.4.2.4 Vibration. Mount the test set on the vibration test fixture as specified in MIL-STD-167-1, type I, except that the following tests shall be performed in lieu of the variable frequency test and endurance test. The equipment shall be verified in the operating condition from a frequency of 5 Hz (or lowest attainable frequency) to 25 Hz in discrete frequency intervals of 1 Hz at the amplitudes shown in table II. The vibration shall be maintained for 1 minute at each integral frequency.

TABLE II. Amplitude-frequency range.

Frequency range (Hz)	Table amplitude single (inch)
5 to 15	0.030 ± 0.006
15 to 25	0.020 ± 0.004

4.4.2.5 Inclination. Mount the test set on the inclination test fixture for the inclination test as specified in MIL-E-16400. Subject the test set to the inclination test as specified therein.

4.4.2.6 Accelerated life. Subject the test set to the accelerated life tests specified in MIL-E-16400. Degradation of performance below specified requirements during this test, or inability to pass the quality conformance test subsequent to this test, shall constitute a failure.

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4.4.3 First article test equipment. The following items of test equipment, or equivalent items, will be required to perform the tests specified in 4.4.2.

a. Temperature-humidity chamber that can provide environments as specified in 4.4.2.1, 4.4.2.2, and 4.4.2.6
b. Shock test fixture for applying shock as specified in 4.4.2.3
c. Vibration test fixture for applying vibration as specified in 4.4.2.4
d. Inclination test fixture for providing inclination attitudes as specified in 4.4.2.5
e. Equipment listed in 4.5.3, quality conformance test equipment.

4.5 Quality conformance test and inspection.

4.5.1 Quality conformance periodic production sampling. When specified (see 6.2), periodic production tests shall be performed at a facility designated by the contracting activity. Samples shall consist of units that have passed the quality conformance inspection (4.5.4 through 4.6.4). The selection shall be as follows:

a. Step 1. Upon initiation of production, randomly select one unit from each group of 20 units. Upon satisfactory compliance of samples selected from three consecutive groups, proceed to step 2.
b. Step 2. Randomly select one unit from each group of 50 units. Continue with step 2 for the balance of the contract, or until a defective unit is found.

4.5.1.1 Quality conformance periodic production tests. Each periodic production sample, after satisfactorily passing the quality conformance tests and examination (4.6), shall be subjected to environmental tests for temperature (4.4.2.1) and for vibration (4.4.2.4). Unless otherwise specified (see 6.2), inability of the sample to satisfactorily pass the quality conformance tests and examination, subsequent to these environmental temperature and vibration tests, shall be deemed a defective unit.

4.5.1.2 Defective units. When a defective unit is found in any group the contractor shall:

a. Determine the cause of the defect
b. Determine if the defect is an isolated case
c. Take the necessary action to eliminate the defect from units on hand, and if so directed by the contracting activity, from those units previously shipped
d. Submit to the contracting activity for approval, the proposed corrective action to prevent recurrence of the defect
e. Return to step 1 (4.5.1 a.) and proceed as indicated for inspection of units subsequently produced.

4.5.2 Test reports. When specified in the contract or purchase order (see 6.2), a test report shall be prepared (see 6.2.1) covering the results of tests specified in 4.4 and 4.5 and forwarded to the contracting activity (see 6.2.1.1).

4.5.3 Quality conformance periodic production test equipment. The following items of test equipment, or equivalent, will be required to perform the periodic production tests:

a. Temperature chamber that can provide an environment as specified in 4.4.2.1
b. Vibration test fixture 4.4.3 c.
c. Equipment as listed in 4.5.4.2.1 quality conformance test equipment.

4.5.4 Quality conformance sampling. The quality conformance tests and inspections (4.5.4.1) shall be conducted on each test set to be delivered; therefore, there is no sampling plan for acceptance. Defective units shall be rejected.

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4.5.4.1 Quality conformance inspection. The following tests and inspections (4.5.4.2 through 4.6.4) shall be conducted on each test set to be delivered. Failure to respond as specified herein shall be considered cause for rejection of the unit.

4.5.4.2 Quality conformance test setup. Interconnect the applicable equipment of 4.5.4.2.1 to the test set as shown in figure 1.

4.5.4.2.1 Quality conformance test equipment. The following items of test equipment or equivalent are required to perform the quality conformance tests of 4.5.4.

Item	*Identification*
a. Resistor (2), 2.5 ohm ±1%, ½W	Commercial
b. Resistor, 2.5 ohm ±1%, ½W	Commercial
c. Resistor (2), 250.0 ohm ±1%, ½W	Commercial
d. Resistor (4), 2500.0 ohm ±1%, ½W	Commercial
e. Connector to mate with J1	Winchester Connector MRE-14P-C7
f. Milliampere current: 0 to 10 milliamperes dc, accuracy ±3% full scale	Commercial

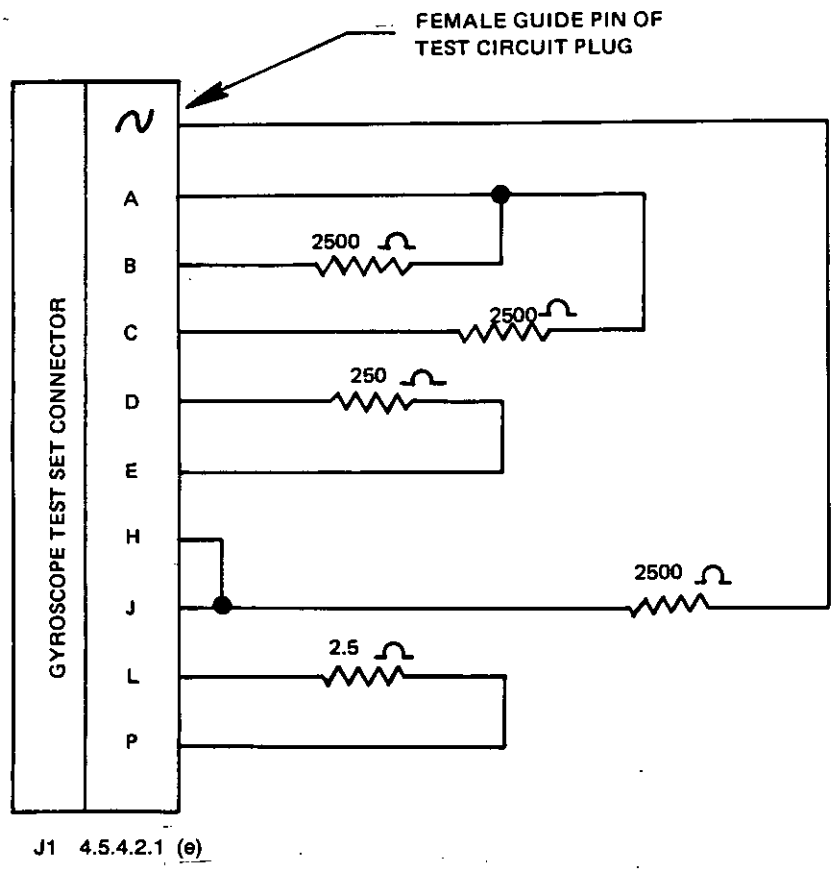


FIGURE 1. Test circuit.

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4.6 Performance tests.

4.6.1 Calibration. Position the test selector switch to calibration, depress the test button, and adjust the calibration adjust control unit until the needle of the ohmmeter coincides with the infinity mark on the scale.

4.6.2 Resistance measurements through connector J1. Rotate the test selector to each position listed in table III. At each position of the test selector, depress the test button. Failure of the ohmmeter to indicate the resistance specified in table III constitutes failure of the test set.

TABLE III. Test selector positions.

Position	Ohmmeter readings
Ins	2500 ± 250
Squib	2.5 ± 0.25
Cont	0.2 maximum
Motor	250 ± 25
Pot	2500 ± 250
Pot	2500 ± 250

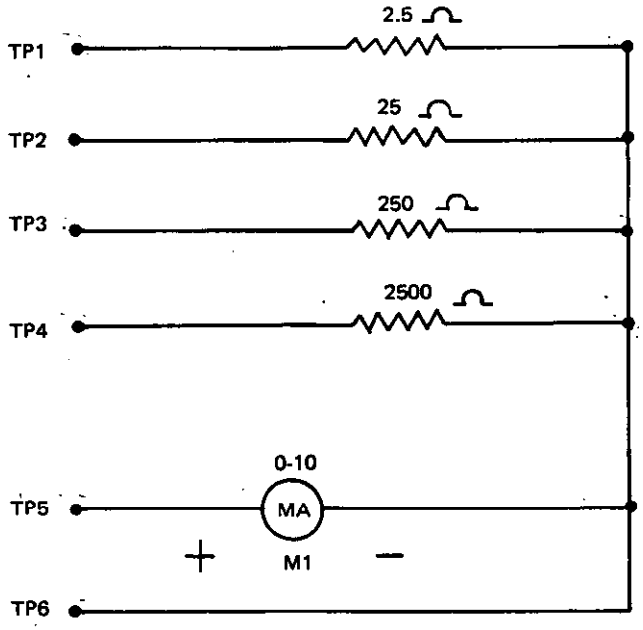
4.6.3 Resistance measurement through test plugs. Position the test selector to ext. Connect P1 (red test plug) to TP6 (see figure 2). Conduct the following tests. Failure of the ohmmeter to indicate a resistance as specified constitutes a failure of the test set.

a. Position the ohms center scale switch to 2.5. Connect P2 (black test plug) to TP1. Depress the test button. The ohmmeter shall indicate 2.5 ± 0.25 ohms.
b. Position the ohms center scale switch to 25. Disconnect P2 from TP1. Connect P2 to TP2. Depress the test button. The ohmmeter shall indicate 25 ± 2.5 ohms.
c. Position the ohms center scale switch to 250. Disconnect P2 from TP2. Connect P2 to TP3. Depress the test button. The ohmmeter shall indicate 250 ± 25 ohms.
d. Position the ohms center scale switch to 2500. Disconnect P2 from TP3. Connect P2 to TP4. Depress the test button. The ohmmeter shall indicate 2500 ± 250 ohms.
e. With ohms center scale switch positioned at 2500, rotate the calibration adjust control fully clockwise. Disconnect P2 from TP4. Connect P2 to TP5. Depress the test button. The milliammeter (see figure 2) shall indicate a current not greater than 10 milliamperes.

4.6.4 Examination of product. The test set shall be examined for conformance to the requirements specified herein and on the drawings listed in PL3277490. The presence of any one of the following defects shall constitute a failure.

a. Test set not complete
b. Parts not new; evidence of reclaimed, used or salvaged parts
c. Evidence of unauthorized materials or repairs
d. Receptacles do not accept mating plugs
e. Dents, scrapes, nicks, bent pins or connections, or any other damage as determined by the contracting activity
f. Panel inscriptions incomplete, incorrect, or illegible
g. Case not watertight
h. Continuity not in accordance with wiring diagram
i. Test set not correctly identified
j. Test set not preserved and packaged in accordance with section 5.

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NOTE: ALL RESISTORS ±1%, 1/2 W

FIGURE 2. Test circuit.

5. PACKAGING

5.1 Preservation and packaging. Preservation and packaging shall be level A, C, or industrial, as specified (see 6.5). Unit packages shall consist of one test set.

5.1.1 Level A. Preservation and packaging of equipment for level A shipment shall be in accordance with MIL-P-116. Cleaning shall be accomplished in accordance with process C-1 and drying, if required, shall be accomplished with method D-1. The equipment case shall serve as a rigid metal container (see 6.5).

5.1.2 Level C or industrial. Cleaning, drying, preservation and packaging shall be accomplished in accordance with the contractor's own commercial practice when such practices have been approved by the contracting activity (see 6.5).

5.2 Packing. Packing shall be level A, C, or industrial, as specified (see 6.2). Shipping container shall contain one unit package.

5.2.1 Level A. The equipment, when preserved and packaged in accordance with 5.1.1, shall be overpacked in containers complying with PPP-B-636, type CF, class — weather resistant, variety SW, Grade V3C (see 5.2.4).

5.2.2 Level C or industrial. The equipment, when preserved and packaged in accordance with 5.1.2, shall be overpacked in compliance with Uniform Freight Classification Rules or other common carrier regulations applicable to the mode of transportation.

5.2.3 Cushioning or dunnage. Contractor shall provide sufficient cushioning or dunnage material to adequately protect the test set from damage due to shock and vibration loads encountered during handling and transportation.

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5.4 Marking. In addition to the marking required by contract or purchase order, unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The Gyroscope Test Set Mark 484 Mod 1 is used to verify assembly of the Course Gyroscope in accordance with the documents listed in LD541118 and PL3277490. Course Gyroscope LD541118 is used in Torpedo Mk 46 Mods 0, 1, and 4; Course Gyroscope, dwg 2557813 is used in Torpedo Mk 46 Mods 2 and 5. Leads from the test set permit measurement of circuit resistances requiring a low test circuit.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification
b. Issue of DODISS to be cited in the soliciation, and if required, the specific issue of individual documents referenced (see 2.1.1, 2.1.2, and 2.2)
c. Serialization requirements
d. Number of test sets required
e. Material and treatment (3.2.1)
f. Environmental test condition (see 3.4 and 4.3)
g. Assigned activity for first article and periodic tests (see 4.1 and 4.4.1)
h. Is periodic sampling test required (see 4.5.1)
i. All additional samples required shall be provided and inspected at the expense of the contractor, when such additional samples are required because of failure of any test sets to pass the prescribed periodic production inspections (see 4.5.1.1)
j. Is test report required (see 4.5.2)
k. Any exceptions to this specification.

6.2.1 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Description (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and the the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractural application of the data requirements, a Contract Data Requirements List (DD Form 1423), must be prepared to obtain the data, except where DOD FAR Supplement, 27.475-1 exempts the requirement for a DD Form 1423.

Reference paragraph	DID Number	DID Title	Suggested Tailoring
4.5.2	DI-T-2072	Reports, Test	

(The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.2.1.1 The data requirements of 6.2.1 and any task in sections 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

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6.3 First article. When a first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.4. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Definitions.

6.4.1 Humidity. The humidity is the relative humidity of the environment to which the test set is exposed, expressed as a percent.

6.4.2 Shock. Shock is an abrupt change in the motion of a body resulting from the sudden application of a large external force of short duration. Shock is described as a pulse in multiples of gravitational acceleration, duration in units of time (milliseconds[ms]), and a shock pulse approximating a one-half sine wave. The time duration shall be measured between two points where two lines, tangent at maximum slope, intersect the zero gravitation line.

6.4.3 Vibration. Vibration is a sinusoidal motion defined by amplitude and frequency. The amplitude may refer to displacement or to acceleration. The displacement is given in inches from a position of rest to peak. The acceleration is given in multiple of gravitational acceleration. The frequency is given in Hertz (Hz).

6.5 Preparation for delivery criteria. Criteria for use of the proper level of preservation, packaging, and packing shall be as follows:

Level A — Maximum protection, designated as level A, is the level of preservation or packing required for protection of material against the most severe worldwide shipment, handling, and storage conditions. Preservation and packing so designated will be designed to protect material against direct exposure to extremes of climate, terrain, operational and transportation environments without protection other than that provided by the pack. The conditions to be considered include, but are not limited to:

a. Multiple handling during transportation and in-transit storage from point of orgin to ultimate user
b. Shock, vibration and static loading during shipment
c. Loading on shipdeck, transfer at sea, helicopter delivery and offshore or over-the-beach discharge, to ultimate user
d. Environmental exposure during shipment or during in-transit operations where port and warehouse facilities are limited or nonexistent
e. Outdoor storage in all climatic conditions for a minimum of 1 year
f. Static loads imposed by stacking.

Level C — Minimum protection designated as level C, will be used for protection of material under known favorable conditions. In general, the following criteria determine the requirements for this level of protection.

a. Use or consumption of the item at the first destination
b. Shock, vibration, and static loading during the limited transportation cycle

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c. Favorable warehouse environmental for a maximum of 18 months
d. Effects of environmental exposure during shipment and in-transit delays
e. Stacking and supporting superimposed loads during shipment and temporary storage.

When level C is specified, it must reference applicable Public Laws (Code of Federal Regulations) or a specific Federal or Military Specification, Standard, or Instruction.

Industrial Packaging. Industrial packaging will be acceptable for any level of protection whenever the technical design details of the package meet all conditions of the level of protection specified. Industrial packaging must provide the same level of protection against physical and environmental damage as the military package. It will be marked to the level it meets. Bulk type practices such as are used in inter and intraplant shipments or shipments to jobbers are not acceptable, unless they are the usual trade practices for selected commodities.

6.6 Subject term (key word listing).

Inch-pound
Test equipment
Torpedo
Revision

6.7 Part number or National Stock Number (NSN).

Part number 3277490
National Stock Number (NSN) 4921-01-066-1760

6.8 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - OS

Project No. 4921-N003

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